

LAB 2C: Which song plays next?**Response Sheet**

Directions: Record your responses to the lab questions in the space provided.

A new direction**Estimate what ... ?**

- (1) Why do we *put a song back* each time we make a selection?

- (2) What would happen in our little experiment if we did not do this?

Calculating probabilities**Estimating probabilities****Getting ready**

```
rap <- rep("rap", times = 39)
```

- (3) Write and run a similar line of code to create the remaining rock songs in our playlist of 100.

Put the songs in the playlist

- (4) Fill in the blanks to combine your different songs:
`songs <- _____(rap, _____)`

Pick a song, any song

- (5) Run this code 10 times and compute the proportion of "rap" songs you drew from the 10.

- (6) Once everyone in your class has computed their *proportions*, calculate the *range of proportions* (the largest *proportion* minus the smallest *proportion*) for your class and write it down.

Now do() it some more

- (7) Fill in the blanks below to do the sample code from the previous slide 50 times:
`do(_____) * sample(_____, _____ = _____, _____ = _____)`

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- (8) Write and run code *assigning* the 50 selected songs the name draws and then View your file.
- (9) What is the variable name?
- (10) Fill in the blank below to tally how often each genre was selected:
`tally(~_____, data = draws)`
- (11) Compute the proportion of "rap" songs for your 50 draws and find out if the *range* for your class's proportions is bigger or smaller than when we drew 10 songs.

Proportions vs. Probability**Non-random Randomness****Playing with seeds**

- (12) What value of `set.seed` did you and your partner use and what was the proportion of "rap" songs you obtained?

Redo the 50 samples one last time but have each partner choose a different number for `set.seed(_____)`.

- (13) Are the proportions still the same? If so, can you find two different values for `set.seed` that give different answers?

On your own

- (14) Write and run code estimating the probability that a randomly chosen student went to the movies using 500 repeated samples.
- (15) Write down both the code and the output that estimates the probability that a randomly chosen student went to the movies using 500 repeated samples. You might find it helpful to write your answer in an R Script (*File -> New File -> R Script*).